



Review Paper

Prevalence and time trends in diabetes and physical inactivity among adult West African populations: The epidemic has arrived

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SUMMARY

Objective: To determine the prevalence and distribution of, and trends in, physical inactivity and diabetes in adult West African populations.**Design:** Systematic review and meta-analysis.**Methods:** Literature searches were conducted using four electronic databases. Journal hand searches and examination of citations of relevant articles were also undertaken. To be included, studies had to be population based, use clearly defined criteria for measuring diabetes and physical inactivity, present data that allowed calculation of the prevalence of diabetes or physical inactivity, and sample adult participants. Studies retrieved were appraised critically. Meta-analysis was performed using the DerSimonian-Laird random effect model.**Results:** Twenty-one reports were retrieved for diabetes and 15 reports were retrieved for physical inactivity. Most studies (10 for diabetes and six for physical activity) were conducted solely among urban populations. The prevalence of diabetes in West Africa was approximately 4.0% [95% confidence interval (CI) 2.0–9.0] in urban adults and 2.6% (95%CI 1.5–4.4) in rural adults, and was similar in men and women [prevalence ratio (PR) 1.36, 95%CI 0.96–1.92]. Cumulative time trend analyses suggested an increase in the prevalence of diabetes among adults in urban West Africa, from approximately 3.0% (95%CI 1.0–7.0) to 4.0% (95%CI 2.0–9.0) in the past 10 years. The prevalence of inactivity in West Africa was 13% (95%CI 9.0–18.0). An association was found between physical inactivity and being older (≥ 50 years) (PR 1.82, 95%CI 1.36–2.44), female gender (PR 1.62, 95%CI 1.41–1.87) and urban residence (PR 2.04, 95%CI 1.58–2.63).**Conclusions:** Diabetes and physical inactivity are important public health issues in urban West Africa, with similar prevalences to wealthy industrialized countries. There is an urgent need for policy makers, politicians and health promotion experts to put measures in place to encourage active lifestyles and control diabetes in urban West Africa.

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Introduction

Diabetes is a major concern for both the developed and the developing world.¹ Approximately 250 million people suffer from the condition worldwide, with up to 10 million of them in sub-Saharan Africa.² The prevalence of diabetes is increasing rapidly, and developing countries, including Africa, will bear the brunt of these increases.¹ Factors responsible for the rise in diabetes include unhealthy diet, overweight/obesity and physical inactivity.³ The World Health Organization (WHO) estimates that 41% of the global population are insufficiently active, and up to 60% fail to achieve the

recommended 30 minutes of moderate activity for most days of the week.³

Physical activity or fitness is associated with reduced all-cause (30–40%), cardiovascular- and cancer-related deaths among chronic disease patients and healthy individuals.^{4,5} Among diabetic patients, people who are less fit (compared with the fit) have twice the risk of death from all causes.⁶ Being physically active improves insulin sensitivity and glucose metabolism, and delays progression to diabetes.⁷

Whilst the definition and diagnostic criteria for diabetes have long been agreed internationally (albeit changed over time), those for physical activity are still under intense debate.^{8,9} Physical activity can be estimated by objective or self-report measures. Self-report measures (questionnaires) have limitations such as social desirability bias, recall bias, subjectivity of responses and

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Table 1

Characteristics and contextual details of included studies for diabetes by country and date of fieldwork or, if not known, publication date.

Study characteristics			Population characteristics			
a. Author	Aim of study and glucose intolerance measure reported	a. Sampling method	Sample size (n)			a. Name of place
b. Date of publication		b. Response rate (%)	Total	Men	Women	b. Rural/urban
c. Date of fieldwork		c. Study design				a. Target population/ sampling frame
Nigeria						
a. Ohwovoriole ¹⁸ b. 1988 c. 1985	Prevalence of diabetes a. Diabetes	a. Census b. NR c. Cross-sectional	1617	1042	575	a. Lagos b. Urban
a. Erasmus ^{19,a} b. 1989 c. NR	Prevalence of diabetes a. Diabetes	a. Census b. NR c. Cross-sectional	2800	1727	1073	a. Illorin b. Urban/rural
a. NEC on Non-Communicable Diseases ^{20,a} b. 1997 c. 1990	Nationwide study of NCD and risk factors	a. Cluster b. 80 c. Cross-sectional	15,064	7857	7207	a. National b. Urban/rural
a. Rotimi ²¹ b. 1999 c. 1994	Prevalence of diabetes a. Diabetes	a. Census b. NR c. Cross-sectional	856	493	363	a. Ibadan b. Urban
a. Owoaje ²² b. 1997 c. 1996	Prevalence of diabetes a. Diabetes	a. Random b. NR c. Cross-sectional	247	108	139	a. Idikan b. Urban
a. Ezenwaka ²³ b. 1997 c. NR	Prevalence of CVD risk factors a. Diabetes	a. Stratified b. NR c. Cross-sectional	500	295	205	a. Idikan and Ibgo-Ora b. Urban/rural
a. Olatunbosun ²⁴ b. 1998 c. NR	Prevalence of diabetes a. Diabetes	a. Random b. NR c. Cross-sectional	875	502	375	a. Ibadan b. Urban
a. Okesina ²⁵ b. 1999 c. 1998	Prevalence of CVD risk factors a. Diabetes	a. Random b. NR c. Cross-sectional	500	278	222	a. Maiduguri b. Rural
a. Nyenwe ²⁶ b. 2003 c. 2000	Prevalence of type 2 diabetes a. Diabetes	a. Cluster b. 67.1 c. Cross-sectional	502	273	229	a. Port Harcourt b. Urban
a. Ogunkolo ²⁷ b. 2005 c. NR	Prevalence of diabetes a. Diabetes	a. Census b. NR (100) c. Cross-sectional	2421	1374	1047	a. Ago-Iwoye b. Urban
Cameroon						
a. Mbanya ^{29,a} b. 1997 c. 1994	Prevalence of diabetes a. Diabetes	a. Census b. 95 c. Cross-sectional	719	292	427	a. Evodouala b. Rural
a. Mbanya ^{29,a} b. 1997 c. 1994	Prevalence of diabetes a. Diabetes	a. Census b. 95 c. Cross-sectional	1048	458	590	a. Yaounde b. Urban
a. Sobngwi ³⁰ b. 2002 c. 1998	Investigate relationship of physical activity and diabetes a. Diabetes	a. Randomly sampled households b. 97 c. Cross-sectional	1183	525	658	a. Biyem-Assi, (Bafut) b. Urban
a. Sobngwi ³⁰ b. 2002 c. 1998	Investigate relationship of physical activity and diabetes a. Diabetes	a. Randomly sampled households b. 97 c. Cross-sectional	1282	523	759	a. Biyem-Assi, Bafut b. Rural
a. Mbanya ^{28,a} b. 2003 c. 2000	Poverty and NCD a. Diabetes	a. Cluster random b. 78.7 c. Cross-sectional	3669	1641	2028	a. Biyem-Assi b. Urban

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Table 1 (continued)

Study characteristics			Population characteristics				
a. Author	Aim of study and glucose intolerance measure reported	a. Sampling method	Sample size (n)			a. Name of place	a. Target population/sampling frame
b. Date of publication		b. Response rate (%)	Total	Men	Women	b. Rural/urban	b. Age range (years)
c. Date of fieldwork		c. Study design					
a. HoPit ³¹ c. 2003 b. 2005	Prevalence of NCD risk factors a. Diabetes	a. Cluster b. NR c. Cross-sectional	9377	3718	5659	a. Multi-city site b. Urban	a. Adults b. ≥15
<i>Ghana</i>							
a. Dodu ³² b. 1964 c. 1963	Prevalence of diabetes a. Diabetes	a. Census b. NR c. Cross-sectional	5537	5537	NA	a. Ho b. Urban	a. Male adults b. ≥15
a. Amoah ³³ b. 2002 c. 1998	Prevalence of diabetes a. Diabetes	a. Cluster b. 75 c. Cross-sectional	4733	1860	2873	a. Greater Accra b. Urban	a. Adults b. ≥25
<i>Mali</i>							
a. Imperato ³⁴ b. 1976 c. 1970/1974	Prevalence of diabetes a. Diabetes	a. NR b. NR c. Cross-sectional	297	NR	NR	a. Bamako, others b. Urban/rural	a. Adults b. ≥15
a. Fisch ³⁵ b. 1987 c. 1984/1985	Prevalence of diabetes a. Diabetes	a. Cluster/census b. NR c. Cross-sectional	7472	NR	NR	a. Rural Mali b. Rural	a. Rural adults b. ≥15
<i>Togo</i>							
a. Teuscher ³⁶ b. 1987 c. NR	Prevalence of diabetes a. Diabetes	a. Census/random b. NR c. Cross-sectional	1381	–	–	a. Agbave, Kati b. Rural	a. Rural Togolese b. >1
<i>Gambia</i>							
a. van der Sande ^{37,a} b. 2001 c. 1996/1997	Cardiovascular risk factors of obesity and undernutrition a. Diabetes	a. Stratified cluster b. 78.1 c. Cross-sectional	5373	NR	NR	a. Banjul, Farafeni b. Urban/rural	a. Adults in Gambia b. ≥15
<i>Sierra Leone</i>							
a. Ceesay ³⁸ b. 1997 c. 1996	Prevalence of diabetes a. Diabetes	a. Cluster b. NR c. Cross-sectional	501	247	254	a. Bo b. Urban/rural	a. Adults b. ≥16

NR, not reported; NEC, National Expert Committee; OGTT, oral glucose tolerance test; IGT, impaired glucose tolerance; WHO, World Health Organization; HoPit, Health of Population in Transition Research Group–Cameroon; CVD, cardiovascular disease.

^a Additional data from WHO InfoBase.

misclassification bias. For instance, what is termed as ‘moderate intensity’ or ‘vigorous intensity’ physical activity could be interpreted differently by different individuals. Despite these limitations, self-report measures remain the most practical, convenient and affordable measures for epidemiological research.¹⁰

Surveillance data on diabetes and its major modifiable risk factors, obesity and physical inactivity, are rare in developing countries, particularly West Africa.^{11,12} This review is an extension and development of an earlier review on diabetes, overweight/obesity and physical inactivity in Ghanaians and Nigerians.¹¹ In that review only nine publications on diabetes and four publications on physical activity were found, making the conclusions tentative, particularly as the quality of much of the data was poor. In view of the suggestion from that work that the prevalence of diabetes was increasing, and the urgency of the authors’ call for further research, it was considered that a review of the larger number of West African countries (17) was needed and, given more and better data, that meta-analysis was required. Limiting the review to these countries was also motivated by the fact that data from the UK and the USA suggest disturbingly high prevalence of diabetes in populations of West African origin.^{13–15} Therefore, conducting this study in the indigenous populations might provide valuable aetiological clues and possible avenues for prevention. This work includes the authors’ earlier data. This article will only report on diabetes and physical inactivity; the

extended review and meta-analysis on obesity have been published recently.¹⁶

Methods

Search and inclusion criteria

Comprehensive electronic literature searches were undertaken using Medline (1966–March 2007), EMBASE (1980–March 2007) and the WHO Global InfoBase for Chronic Disease Surveys. Searches were also conducted on all available issues of African Journals Online. Details of the methods have been reported elsewhere¹³ (for obesity). Searches in Medline and EMBASE included both medical subject headings (MeSH) and key text. Search terms included diabetes, diabetes mellitus, type 2 diabetes, non-insulin-dependent diabetes, NIDDM, insulin, impaired glucose tolerance, IGT, physical activity, physical inactivity, active, sports, fitness and exercise. Searches were combined with Africa, West Africa and names of all countries in West Africa using appropriate Boolean operatives (‘OR’ or ‘AND’). The following countries were considered as West Africa: Benin, Burkina Faso, Cameroon, Cape Verde, Cote d’Ivoire, Gambia, Ghana, Guinea, Guinea-Bissau, Liberia, Mali, Mauritania, Niger, Nigeria, Senegal, Sierra Leone and Togo. The WHO InfoBase for Chronic Diseases was searched for studies and reports on diabetes and physical in/activity for each of the above countries.

Table 2

Characteristics and contextual details of included studies for physical (in)activity by country and date of fieldwork or, if not known, publication date.

Study characteristics				Population characteristics			
a. Author	Aim of study	Method of assessment	a. Sampling method	Sample size (n)			a. Name of place
b. Date of publication			b. Response rate (%)	Total	Men	Women	b. Rural/urban
c. Date of fieldwork			c. Study design				c. Country
<i>Nigeria</i>							
a. NEC on Non-Communicable Diseases ^{20,a}	Risk factors for cardiovascular disease	Description not available	a. Cluster	15,558	NR	NR	a. National
b. 1990			b. 80				b. Rural/urban
c. 1997			c. Cross-sectional				c. Nigeria
a. Forrest ⁴¹	Evaluate the role of PA in CVD	Average hours per week over the past year spent in occupation and leisure	a. Census	799	498	301	a. Benin City
b. 2000			b. 84				b. Urban
c. 1992			c. Cross-sectional				c. Nigeria
a. Ezenwaka ²³	Prevalence of CVD risk factors	PA assessed by quantitative ranking of daily activities	a. Stratified	500	295	205	a. Idikan and Ibgbo-Ora
b. 1997			b. NR				b. Urban/rural
c. NR			c. Cross-sectional				c. Nigeria
a. Nyenwe ²⁶	Prevalence of type 2 diabetes	Assessed from the occupation of the individual	a. Cluster	502	273	229	a. Port Harcourt
b. 2003			b. 67.1				b. Urban
c. 2000			c. Cross-sectional				c. Nigeria
a. Ojofetimi ⁴²	Assess risk factors of cancer in hospital workers	Self-reported frequency of exercise	a. NR	169	NR	NR	a. Ile-Ife, Osogbo
b. 2003			b. NR				b. Urban
c. 2001/2002			c. Cross-sectional				c. Nigeria
a. Afolabi ⁴³	Establish prevalence of overweight and obesity	Self-reported pattern of activities and lifestyle	a. Systematic random	232	NA	232	a. Markets in Abeokuta
b. 2004			b. 97				b. Urban
c. NR			c. Cross-sectional				c. Nigeria
<i>Cameroon</i>							
a. Sobngwi ³⁰	Evaluate and compare physical activity in urban and rural Cameroon	Habitual physical activity assessed from past year's occupation, walking and leisure-time activities	a. Random sampled households'	965	1360	2325	a. Biyem-Assi, Bafut
b. 2002			b. 97				b. Urban/rural
c. 1998			c. Cross-sectional				c. Cameroon
a. HoPiT ³¹	Baseline survey for diabetes and risk factors	Workplace and leisure-time physical activity	a. Cluster	10,011	4007	6004	a. Multi-centre survey (four cities)
b. 2004			b. NR				b. Urban
c. 2003			c. Cross-sectional				c. Cameroon
a. Kengne ⁴⁴	Burden of high BP and related risk factors	Workplace and leisure-time physical activity	a. Multi-level stratified	2559	1031	1528	a. Douala
b. 2007			b. 92.5				b. Urban
c. 2004			c. Cross-sectional				c. Cameroon
<i>Gambia</i>							
a. van der Sande ^{37,a}	Cardiovascular risk factors of obesity and undernutrition	Spent less than half a day on feet or led sedentary life	a. Stratified cluster	5389	2228	3161	a. Banjul, Farafeni
b. 2001			b. 78.1				b. Urban/rural
c. 1996–1997			c. Cross-sectional				c. Gambia
<i>World Health Surveys</i>							
<i>Burkina Faso</i>							
a. WHO ⁴⁵	World Health Survey (cardiovascular disease risk factor assessment)	International physical activity questionnaire	a. Cluster	2020	2321	4341	a. National
b. NA			b. 95.6				b. Urban/rural
c. 2003			c. Cross-sectional				c. Burkina Faso
<i>Cote d' Ivoire</i>							
a. WHO ⁴⁵	World Health Survey (cardiovascular disease risk factor assessment)	International physical activity questionnaire	a. Cluster	1388	1049	2437	a. Subnational

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Table 2 (continued)

Study characteristics				Population characteristics			
a. Author	Aim of study	Method of assessment	a. Sampling method	Sample size (n)			a. Name of place
b. Date of publication			b. Response rate (%)	Total	Men	Women	b. Rural/urban
c. Date of fieldwork			c. Study design				c. Country
b. NA c. 2003			b. 96.5 c. Cross-sectional				b. Urban/rural c. Cote d'Ivoire
<i>Ghana</i>							
a. WHO ⁴⁵	World Health Survey (cardiovascular disease risk factor assessment)	International physical activity questionnaire	a. Cluster	3362	1532	1830	a. National
b. NA c. 2003			b. 69.6 c. Cross-sectional				b. Urban/rural c. Ghana
<i>Mali</i>							
a. WHO ⁴⁵	World Health Survey (cardiovascular disease risk factor assessment)	International physical activity questionnaire	a. Cluster	899	579	1478	a. National
b. NA c. 2003			b. 78.7 c. Cross-sectional				b. Urban/rural c. Mali
<i>Senegal</i>							
a. WHO ⁴⁵	World Health Survey (cardiovascular disease risk factor assessment)	International physical activity questionnaire	a. Cluster	806	804	1610	a. National
b. NA c. 2003			b. 88.4 c. Cross-sectional				b. Urban/rural c. Senegal

NEC, National Expert Committee; NR, not reported; PA, physical activity; CVD, cardiovascular disease; BP, blood pressure; NA, not applicable.

^a Additional data from WHO InfoBase.

Bibliographic lists of retrieved studies were also searched and, where possible, authors were contacted for further information (see Acknowledgements).

Titles and abstracts were examined for inclusion by two independent reviewers (ARA and MJ) based on predefined inclusion criteria, and any disagreements were resolved by consensus. The predefined criteria for inclusion were: studies had to be population based (i.e. sampled unrestricted/free living men and women drawn from identifiable communities); use well-defined criteria for diabetes or physical in/activity (such as WHO's definition); and present crude data or figures that allowed calculation of the prevalence of diabetes or physical in/activity.

Data from each study were extracted using a form designed to include context and population characteristics (age, urban/rural residence, sample size, sampling method, response rates) and on the outcomes of interest (physical in/activity or diabetes). The extraction form and the data collected were independently reviewed by two reviewers (WL and AK).

Statistical analysis

Data were tabulated by country, gender and rural/urban residence. To examine trends, studies were also tabulated by year of fieldwork. Due to wide variation in methodological approaches, particularly in the definition and diagnostic criteria for diabetes and physical inactivity, meta-analysis was only conducted on studies with similar diagnostic or assessment criteria. For instance, pooled analysis for physical inactivity was only performed on the World Health Survey data, whilst time trend analysis for diabetes was limited to studies that used the WHO's recommended criteria (1985 or later). Prevalence ratios were used for comparisons. For studies in which mean ages were not reported, they were derived using the probability weighted average if the sample size for individual age strata were given. The DerSimonian-Laird random effect model was

used for meta-analysis. Statistical analysis was conducted using programmes in R.¹⁷

Results

Description of context and methods of studies

Diabetes

As shown in Table 1, 21 studies/reports on the prevalence of diabetes in various West African populations were included. Excluded studies, reasons for exclusion and their full references are shown in Appendices 1 and 2. Most included studies were conducted in Nigeria^{18–27} (two more than reviewed previously¹¹: one newly published²⁷ and one from WHO InfoBase²⁰), followed by Cameroon,^{28–31} Ghana,^{32,33} Mali,^{34,35} Togo,³⁶ Gambia³⁷ and Sierra Leone.³⁸ Dates of fieldwork (often missing) ranged from 1963³² to 2003.³¹ Diagnostic criteria varied widely across studies, although most used one of the three sets of WHO-recommended criteria.^{8,39,40} Most studies used various probability^{20,22–27,29,33} or census^{18,19,21,27,29,32} sampling methods. Sample sizes ranged between 247²² and 9377,³¹ and satisfactory to high response rates (67–95%) were reported although this information was frequently missing. Dodu and de Heer³² and Imperato et al.³⁴ only included male subjects in their studies, but the other studies included both male and female subjects. Three studies were conducted exclusively in rural populations,^{25,35,36} 10 were conducted exclusively in urban populations,^{18,21,22,24,26,28,31–33} and the remaining studies were conducted in both urban and rural populations. The age range for participants also varied substantially, although most studies sampled adults through the age range.

Physical in/activity

Subnational studies: Ten subnational population-based and five unpublished nationally representative studies were found (Table 2).

Table 3

Prevalence of diabetes among adult West Africans by gender, year of fieldwork, rural/urban residence and country.

First author	Year of fieldwork	Diagnostic criteria	Sample size	Mean age (SD)		Mean blood glucose (FBG)			Percentage diabetes		
				Men	Women	All	Men	Women	All	Men	Women
<i>Nigeria</i>											
<i>Urban</i>											
Ohwovoriole ¹⁸	1985	WHO 1985 (urine dipstick then capillary blood glucose test; if fasting ≥ 6.6 , if non-fasting ≥ 10.0)	1617	33.7 (14.2)	34.6 (16.7)	NR	NR	NR	1.64 (1.2–2.4)	1.54 (1.0–2.5)	1.91 (1.1–3.4)
Rotimi ²¹	1994	WHO 1980 (FBG ≥ 8.0 or OGTT ≥ 11.0)	856	43.1 (8.1)	37.3 (6.3)	NR	4.4	4.6	0.80 (0.4–1.7)	1.0 (0.4–2.4)	0.6 (0.2–2.0)
Owoaje ²²	1996	WHO 1985 (FBG ≥ 6.7)	247	53.6 (12.6)	50.6 (12.8)	4.9 (1.1)	4.7 (1.3)	4.9 (1.1)	2.8 (1.4–5.7)	2.8 (1.0–7.9)	2.9 (1.1–7.2)
Olatunbosun ²⁴	NR	WHO 1985 (75 g OGTT, plasma glucose ≥ 11.1)	875	42.5 (8.5)	36.6 (6.9)	NR	NR	NR	0.80 (0.4–1.6)	0.80 (0.3–2.0)	0.80 (0.3–2.3)
Nyenwe ²⁶	2000	WHO 1999 (casual plasma glucose ≥ 7.0 then 75 g OGTT ≥ 11.1)	502	50.2 (8.9)	47.5 (9.2)	NR	NR	NR	6.80 (4.9–9.3)	7.70 (5.1–11.5)	5.70 (3.4–9.5)
Ogunkolo ²⁷	NR	ADA 1997 (symptoms + RCBG > 11.1 or FBGC ≥ 7.0)	2421	24.9	22.1	NR	NR	NR	3.50 (2.8–4.3)	2.90 (2.2–3.9)	4.20 (3.2–5.6)
<i>Urban/rural</i>											
Erasmus ^{19,a}	NR	WHO 1980 (FPG if values between 6.5–7.7 then 75 g OGTT)	2800	28.1	24.0	4.6	4.6 (0.8)	4.5 (0.6)	1.43 (1.1–1.9)	1.45 (1.0–2.1)	1.40 (0.9–2.3)
NEC on Non-Communicable Diseases ^{20,a}	1990	(Fasting blood glucose > 6.60 or casual blood glucose > 9.9)	15,064	36.6	36.5	NR	NR	NR	2.80 (2.5–3.1)	2.70 (2.4–3.1)	3.00 (2.6–3.4)
Ezenwaka ²³	NR	WHO 1985 (FPG > 7.8)	500	60.5 (1.2)	61.2 (1.5)	4.3 (0.1)	4.2 (0.1)	4.3 (0.1)	1.60 (0.8–3.1)	1.40 (0.5–3.4)	1.90 (0.8–4.9)
<i>Rural</i>											
Okesina ²⁵	1998	Fasting blood glucose (other details not reported)	500	29.3	24.2	NR	4.3	6.3	2.60 (1.5–4.4)	1.08 (0.4–3.1)	4.50 ^b (2.5–8.1)
<i>Cameroon</i>											
<i>Urban</i>											
Mbanya ^{29,a}	1994	WHO 1985 (75 g OGTT ≥ 11.1)	1048	37	36	NR	4.2	4.1	1.34 (0.8–2.2)	1.10 (0.5–2.5)	1.53 (0.8–2.9)
^c Sobngwi ³⁰	1998	WHO 1999 (fasting capillary glucose ≥ 6.1)	1183	26	27	4.8	4.7 (4.4–5.1)	4.8 (4.5–5.2)	5.4 (4.3–6.9)	6.2 (4.5–8.7)	4.7 (3.3–6.6)
Mbanya ^{28,a}	2000	WHO 1999 (Fasting blood glucose ≥ 6.1)	3669	31.7	31.6	NS	NS	NS	1.7 (1.3–2.2)	2.3 (1.7–3.2)	1.2 (0.8–1.8)
HoPiT ³¹	2003	WHO 1999 (fasting blood glucose ≥ 6.1 or OGTT)	9377	~ 34.5	~ 34.5	NR	NR	NR	5.2 (4.3–5.2)	5.6 (4.4–5.8)	4.9 (4.3–5.4)
<i>Rural</i>											
Mbanya ^{29,a}	1994	WHO 1985 (75 g OGTT ≥ 11.1)	719	43	47	NR	4.0	4.1	0.83 (0.4–1.8)	1.03 (0.4–3.0)	0.70 (0.2–2.1)
^c Sobngwi ³⁰	1998	WHO 1999 (fasting capillary glucose ≥ 6.1)	1282	48	46	4.6	4.6 (4.1–5.1)	4.5 (4.2–5.0)	3.7 (2.8–4.8)	4.7 (3.3–7.0)	2.9 (1.9–4.4)
<i>Ghana</i>											
<i>Urban</i>											
Dodu ³²	1963	Urine dipstick, if glycosuria Positive, then 50 g OGTT	5537	34.0	NA	NR	NR	NR	NA (0.1–0.4)	0.22 (0.1–0.4)	NA
Amoah ³³	1998	WHO 1998 (fasting plasma glucose ≥ 7.0 and /or 75 g OGTT 11.1)	4733	44.9 (14.7)	44.0 (14.6)	5.5 (1.2)	5.5 (1.1)	5.4 (1.3)	6.30 (5.6–7.0)	7.80 (6.7–9.1)	5.40 ^b (4.6–6.3)
<i>Mali</i>											
<i>Urban/rural</i>											
Imperato ³⁴	1970/74	Postprandial blood glucose level > 170 mg/100 ml	297	37.0	35.2	NR	NR	NR	NA (0.5–3.4)	1.40 (0.5–3.4)	NA
<i>Rural</i>											
Fisch ³⁵	1984/85	WHO 1985 (fasting glycaemia ≥ 7.0) ^b Not in line with the above criteria	7472	~ 34.3	~ 34.3	NR	NR	NR	0.92 (0.7–1.2)	0.78 (0.5–1.1)	1.05 (0.8–1.4)
<i>Gambia</i>											
<i>Urban/rural</i>											
van der Sande ^{37,a}	1996/97	Whole blood 2-h OGTT ≥ 10.0 mmol/l	2394	35.1	35.5	NR	NR	NR	3.40 (2.7–4.2)	4.00 (2.9–5.5)	3.00 (2.2–4.0)

(continued on next page)

Table 3 (continued)

First author	Year of fieldwork	Diagnostic criteria	Sample size	Mean age (SD)		Mean blood glucose (FBG)			Percentage diabetes		
				Men	Women	All	Men	Women	All	Men	Women
<i>Togo</i> <i>Rural</i> Teuscher ³⁶	NR	WHO 1985 (random capillary hole blood, >11.0 or 5 g OGTT)	1381	NR	NR	5.1 (1.1) CBG	5.1 (1.0)	5.3 (1.1)	0.00	0.00	0.00
<i>Sierra Leone</i> <i>Urban</i> Ceasey ³⁸	1996	WHO 1985 (random capillary blood glucose ≥ 11.1)	245	33.8 (15.5)	33.9 (17.2)	5.8 (2) RBG	5.6 (2.4)	5.9 (1.7)	2.45 (1.1–5.2)	NR	NR
<i>Rural</i> Ceasey ³⁸	1996	WHO 1985 (random capillary blood glucose ≥ 11.1)	256	36.9 (14.7)	38.1 (14.4)	5.5 (1.4)	5.5 (14.7)	5.6 (1.4)	0.00	0.00	0.00

CBG, capillary blood glucose; FPG, fasting plasma glucose; NR, not reported; RBG, random blood glucose; NEC, National Expert Committee; HoPiT, Health of Population in Transition Research Group–Cameroon; WHO, World Health Organization; ADA, American Diabetes Association.

^a Additional data from WHO InfoBase.

^b Significant difference between men and women.

^c Sobngwi reported standardized prevalence.

All 10 subnational surveys were conducted between 1990 and 2004. Six of the studies reported on Nigerians^{20,23,26,41–43} (three had not been reviewed previously¹¹; two were missed by the earlier search^{42,43} and one was found on WHO InfoBase²⁰), three on Cameroonians^{30,31,44} and one on Gambians.³⁷ Ojofeiti et al.⁴² did not report details of the sampling procedure, but most studies used probability sampling methods. Response rates ranged between 67.1%²⁶ and 97%,⁴³ whilst sample size ranged between 169⁴² and 15,558.²⁰ Six studies were conducted exclusively in urban populations,^{26,41–44} whilst four studies combined results for urban and rural populations. Afolabi et al.⁴³ only included Nigerian market women in their study, but the other studies included both men and women. Ezenwaka et al.²³ and Nyenwe et al.²⁶ studied middle-aged adults (≥ 50 and ≥ 40 years, respectively), but all other studies used wider age ranges (≥ 15 or ≥ 20 years old). Study participants were government department staff,^{41,42} market women⁴³ or adults from the community.^{20,23,26,31,37,44}

Physical activity was measured by self-report questionnaires and the definition for in/activity varied widely across studies. The domains of physical activity assessed included occupational,²⁶ occupation and leisure,^{30,31,41,44} daily activities and lifestyle,^{23,37} or frequency of exercise.⁴² Most studies did not report the duration, intensity and frequency at which participants were deemed active

or otherwise. For studies that reported percentage prevalence of physical activity, this was subtracted from 100 to determine prevalence of inactivity.

World Health Organization's national surveys: Five World Health Surveys were retrieved.⁴⁵ The surveys were conducted in Burkina Faso, Cote d'Ivoire, Ghana, Mali and Senegal in 2002/2003. A similar protocol was used in conducting the surveys which were part of the WHO's Non-communicable Diseases and risk factors assessment involving 72 countries. Participants were aged between 18 and 69 years. Total sample sizes ranged from 1478 in Mali to 4341 in Burkina Faso. Physical activity was measured using the International Physical Activity Questionnaire (IPAQ) which was adapted to suit individual countries. Using IPAQ, physically inactive refers to those who do not meet the criteria for high or moderate physical activity levels. These are: 3 or more days of vigorous activity of at least 20 minutes per day, 5 or more days of moderate intensity activity or walking for at least 30 minutes per day, or 5 or more days of any combination of the above (vigorous intensity, moderate intensity or walking), achieving not less than 600 metabolic equivalents in energy expenditure per week (i.e. the sum product of vigorous, moderate or walking activities over a week).⁹

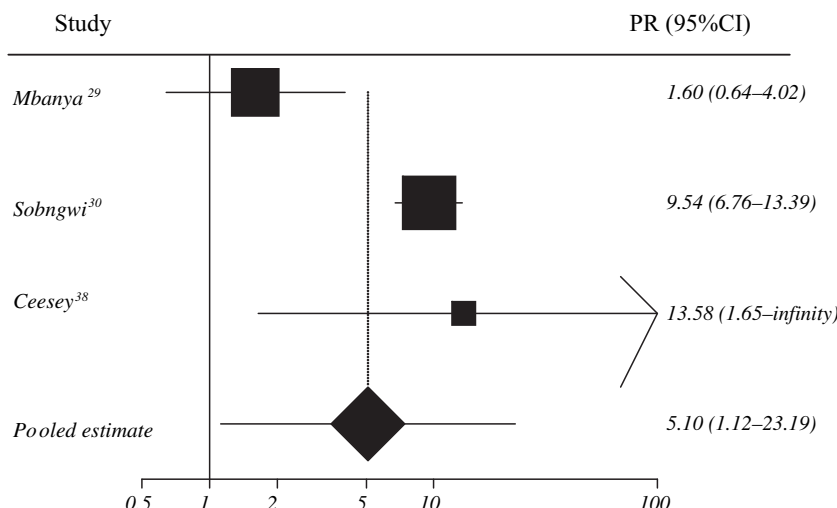


Figure 1. Prevalence ratio (PR) and 95% confidence intervals (CI) of diabetes in urban West Africa compared with rural West Africa.

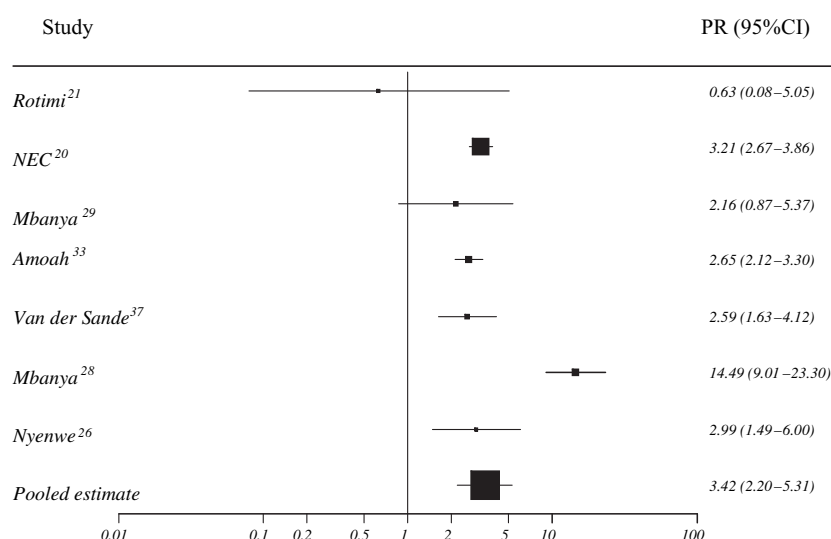


Figure 2. Prevalence ratio (PR) and 95% confidence intervals (CI) of diabetes in older (>50 years) West Africans compared with younger (<50 years) West Africans.

Prevalence of diabetes and physical inactivity

Diabetes

Table 3 shows the prevalence of diabetes in various West African populations as reported by the included studies. The substantial variation in study methods (see Table 1) makes it difficult to interpret the data. However, recent studies have suggested that at least 5% of urban adults for most countries have diabetes.^{26,30,31,33} West Africans residing in urban areas were over five times more likely to have diabetes (Fig. 1) compared with their rural-dwelling counterparts [pooled prevalence ratio (PR) 5.10, 95% confidence interval (CI) 1.12–23.19]. The urban/rural difference was also observed in most individual West African countries, including Cameroon,³⁰ Nigeria²⁵ and Sierra Leone.³⁸ The prevalence of diabetes among men compared with women differed between countries.^{25,28,33} The pooled analysis suggests a gender difference but the confidence intervals include 1 (1990–1999: PR 1.36, 95%CI 0.96–1.92; 2000 onwards: PR 1.40, 95%CI 0.84–1.53). Diabetes was also found to be more prevalent in the elderly than the young.^{18,19,33,35,37} As shown in Fig. 2, diabetes among older West Africans (age ≥50 years) was over three times higher than among younger (age <50 years) West Africans (PR 3.42, 95%CI 2.20–5.31).

Time trends were difficult to discern due to differences in definitions of diabetes (diagnostic criteria), population characteristics, lack of fieldwork dates and lack of data for individual years. However, it was possible to compare studies using the WHO criteria (1985 or later) within some countries, and also across the past decade or two. The results suggest that the prevalence of diabetes increased by more than 300% between 1985 and 2000 in urban Nigeria,^{18,26} and between 1994 and 2003 in urban Cameroon.^{29,31,46} The pooled estimates in Table 4 also suggest that the prevalence of diabetes has increased by approximately 30% in the last decade [proportion difference between 1991–1999 and 2000 onwards 33%, (95%CI 15.33–50.66, $\chi^2 = 12.18$, $P < 0.001$). However, as illustrated in Fig. 3, the increase in the prevalence of diabetes was higher and significant in men (χ^2 for linear trend 37.67, $P < 0.0001$) compared with women (χ^2 for linear trend 0.84, $P = 0.36$).

Physical inactivity

Table 2 shows the wide variation in definitions of physical inactivity, making it difficult to compare the results of the published studies. This notwithstanding, the data from specific studies indicated that physical inactivity could be higher than expected in this population. Three-quarters of urban West African populations

Table 4

Reports examining prevalence of diabetes in urban West Africa using the World Health Organization recommended criteria (1985 or later) by gender and year of fieldwork.

Author	Year of fieldwork	Age structure (mean age if not age range)			Percentage diabetes		
		All	Men	Women	All	Men	Women
<i>Before 1990</i>							
Ohwovoriole ¹⁸	1985	34.2	33.7 (14.2)	34.6 (16.7)	1.64 (1.2–2.4)	1.54 (1.0–2.5)	1.91 (1.1–3.4)
<i>1991–1999</i>							
Mbanya ²⁹	1994	36.5	37.0	36.0	1.34 (0.8–2.2)	1.10 (0.5–2.5)	1.53 (0.8–2.9)
Owoaje ²²	1996	52.1	53.6 (12.6)	50.6 (12.8)	2.80 (1.4–5.7)	2.80 (1.0–7.9)	2.90 (1.1–7.2)
Ceesay ³⁸	1996	33.9 (16.5)	33.8 (15.5)	33.9 (17.2)	2.45 (1.1–5.2)	NR	NR
Amoah ³³	1998	44.3 (14.7)	44.9 (14.7)	44.0 (14.6)	6.30 (5.6–7.0)	7.80 (6.7–9.1)	5.40 (4.6–6.3)
Pooled prevalence					3.00 (1.0–7.0)	3.00 (1.0–13.0)	3.00 (1.0–8.0)
<i>2000 onwards</i>							
Nyenwe ²⁶	2000	48.9 (9.2)	50.2 (8.9)	47.5 (9.2)	6.80 (4.9–9.3)	7.70 (5.1–11.5)	5.70 (3.4–9.5)
Mbanya ²⁸	2000	43.0	31.7	31.6	1.70 (1.3–2.2)	2.30 (1.7–3.2)	1.20 (0.8–1.8)
HoPiT ³¹	2003	34.5	~34.5	~34.5	5.20 (4.3–5.2)	5.60 (4.4–5.8)	4.90 (4.3–5.4)
Pooled prevalence					4.00 (2.0–9.0)	5.00 (3.0–9.0)	3.00 (1.0–9.0)

NR, not reported.

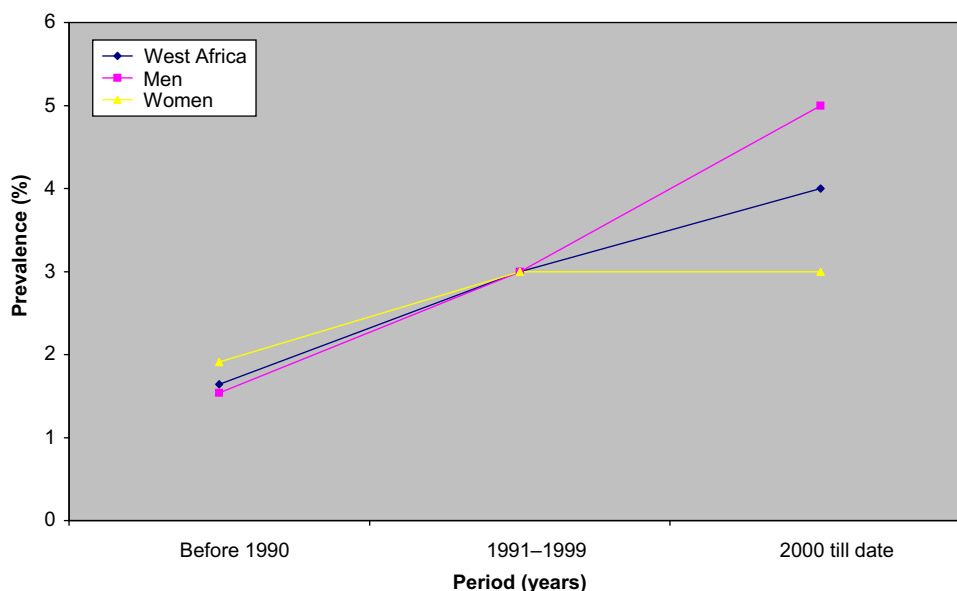


Figure 3. Prevalence of diabetes in urban West African adults (data from studies in Table 4). Note: Chi-square and *P*-values for linear change in diabetes trend between 1985 and post 2000 as in Figure 3 are: West Africa: $\chi^2 = 30.37$, $P < 0.0001$; men: $\chi^2 = 37.67$, $P < 0.0001$; women: $\chi^2 = 0.84$, $P = 0.36$.

were described as sedentary in their occupations,^{30,31} and half of public sector workers did not undertake any form of exercise.⁴²

The World Health Survey data (Table 5) indicated that 7–19% of populations in West African countries were inactive; the range indicates the wide variation between countries. For instance, the prevalence of inactivity in urban Burkinabe women was 11% compared with >30% among their Malian counterparts. Inactivity was higher among women compared with men, and in urban residents compared with rural residents. Age-specific prevalence of inactivity increased with age in all countries.

The pooled analysis estimated that 13% (95%CI 9.0–18.0) of West African populations were inactive. The pooled inactivity in West African women was higher than in men (PR 1.62, 95%CI 1.41–1.87). Estimated combined inactivity was 18% (95%CI 14.0–22.0) in urban populations and 9.0% (95%CI 6.0–13.0) in rural populations. Overall, age-specific prevalence increased from 12% in 18–29 year olds to 30% in 60–69 year olds. The age-specific trend was the same irrespective of gender or place of residence (Fig. 4). Thus, women (PR 1.62, 95%CI 1.41–1.87), people aged ≥ 50 years (PR 1.82, 95%CI 1.36–2.44) or those residing in an urban community (PR 2.04, 95%CI 1.58–2.63) were more likely to be inactive than their male, younger (<50 years) or rural counterparts, respectively.

Discussion

Principal findings

Surveillance data for non-communicable diseases, including diabetes and its risk factors, in West African countries are rare. In 17 countries, only 21 studies reported on diabetes and 15 studies reported on physical activity, despite the fact that the selection criteria were not stringent. This situation hinders effective design, planning and monitoring of disease control and prevention services.

It is estimated that 4% of urban West African adults have diabetes, and this figure is higher in some countries. The prevalence is similar in men and women, although the prevalence of obesity is much higher in women than men.¹⁶ The disease is particularly common among urban residents and increases with age. The

prevalence of diabetes has increased greatly, especially in urban areas, over the past few decades and continues to rise.

One in seven West Africans are physically inactive. Some studies report that up to three-quarters of urban residents are sedentary in their daily occupations,^{30,31} and half of government workers do not undertake any leisure-time physical activity.⁴² Inactivity is higher among women compared with men, and among urban residents compared with rural residents.

Discussion of principal findings

The demographic structures and disease patterns in West African populations are in transition. The disease burden is thereby shifting from mainly communicable to both communicable and chronic non-communicable diseases, hence the 'double burden of disease'.⁴⁷ The high prevalence of diabetes in urban West Africa is a public health issue of concern. Diabetes was virtually non-existent in West African populations about four decades ago.^{32,34,36} This partially reflects the ageing population structure. However, urbanization and changing diet and lifestyles are probably playing the lead role for the observed epidemic.⁴⁸ Urbanization leads to increased consumption of diets high in fat and sugar, that are characterised with high caloric content. Urbanization is also accompanied by a shift from labour-intensive occupational activities to more sedentary, service-based occupations that require less energy. This observation is reflected in the physical inactivity of urban West Africans seen in this review. The population, however, is failing to engage in leisure-time physical activities to compensate for the huge reduction in energy expenditure through occupational activity.^{31,44} Recent surveys in a number of Cameroonian cities, for instance, indicate that over 70% of urban residents are sedentary in their workplace, and over 90% of these people are also inactive during their leisure time.^{30,31} The available data do not permit the examination of trends in diabetes, obesity and physical inactivity simultaneously in this population, yet the higher prevalence of diabetes in the urban centres surely reflects the distribution of the classical risk factors (obesity and physical inactivity). Most of the studies included in this review also reported significant cross-sectional associations between diabetes and physical inactivity and between diabetes and obesity or body mass index in this

Table 5

Prevalence of physical inactivity in adult West Africans by year of fieldwork, gender and country.

Study	Survey year	Mean age			Percentage inactive	
Country		Men	Women	All	Men	Women
<i>Nigeria</i>						
<i>Urban</i>						
Forest ⁴¹ (biking/walking)	1992	42.5	39.0	25.30 (22.2–28.4)	27.00 (23.4–31.2)	22.60 (18.2–27.6)
Nyenwe ²⁶	2000	50.2 (8.9)	47.5 (9.2)	40.10 (35.9–44.4)	NR	NR
Ojofetimi ⁴²	2001/2002	NR	NR	50.00 (42.8–57.8)	NR	NR
Afolabi ⁴³	NR	NA	40.3 (12.5)	45.00 (38.6–51.3)	NA	45.00 (38.6–51.3)
<i>Rural/urban</i>						
NEC on Non-Communicable Diseases ^{20,a}	1990	36.6	36.5	6.80 (6.4–7.2)	NR	NR
Ezenwaka ²³	NR	60.5 (1.2)	61.2 (1.5)	57.20 (52.8–61.5)	39.30 (33.9–45.0)	82.90 (77.2–87.5) [‡]
<i>Cameroon</i>						
<i>Urban</i>						
Sobngwi ³⁰	1998	26.0	27.0	69.40 (66.7–72.1)	68.20 (63.9–72.3)	70.20 (66.5–73.7)
HoPiT ³¹	2003	~34.5	~34.5	73.0 (72.1–73.9)	NR	NR
Kengne ⁴⁴	2004	NR	NR	79.6 (78.0–81.1)	NR	NR
<i>Rural</i>						
Sobngwi ³⁰	1998	48	46	6.9 (5.6–8.4)	9.6	5.0 (3.7–6.9)
<i>Gambia</i>						
<i>Urban/rural</i>						
van der Sande ^{37,a}	96/97	35.1	35.5	56.58 (55.3–57.9)	48.60 (46.5–50.7)	62.30 (60.6–64.0) [‡]
<i>World Health Surveys⁴⁵</i>						
<i>Burkina Faso</i>						
<i>Urban</i>						
	2003	35.7	33.5	11.30 (8.5–14.0)	11.10 (7.0–15.3)	11.40 (7.9–14.9)
<i>Rural</i>						
	2003	36.5	34.2	5.70 (4.4–7.0)	4.50 (3.0–6.1)	6.70 (5.0–8.4)*
<i>Urban/rural</i>						
	2003	36.2	34.0	6.70 (5.5–7.9)	5.80 (4.3–7.4)	7.40 (5.9–9.0)*
<i>Cote d'Ivoire</i>						
<i>Urban</i>						
	2003	34.0	33.0	17.10 (13.9–20.4)	12.10 (9.2–15.1)	23.80 (18.6–29.0) [‡]
<i>Rural</i>						
	2003	35.9	36.0	5.90 (3.6–8.3)	5.80 (3.2–8.3)	6.20 (3.1–9.3)
<i>Urban/rural</i>						
	2003	34.7	34.0	13.90 (11.4–16.4)	10.30 (8.1–12.5)	18.80 (14.8–22.8) [‡]
<i>Ghana</i>						
<i>Urban</i>						
	2003	37.2	38.5	16.50 (12.9–20.1)	11.70 (7.8–15.6)	21.10 (16.6–25.7) [‡]
<i>Rural</i>						
	2003	39.0	39.1	7.30 (6.0–8.6)	4.60 (3.1–6.1)	9.90 (7.9–12.0) [‡]
<i>Urban/rural</i>						
	2003	38.3	38.8	11.50 (9.7–13.3)	7.90 (5.9–9.8)	15.10 (12.7–17.5) [‡]
<i>Mali</i>						
<i>Urban</i>						
	2003	37.7	36.4	25.00 (17.0–33.0)	18.30 (8.7–27.9)	30.90 (20.1–41.8)
<i>Rural</i>						
	2003	39.1	37.8	11.60 (9.1–14.0)	9.20 (6.8–11.6)	14.20 (10.4–18.0)*
<i>Urban/rural</i>						
	2003	38.9	37.5	15.20 (12.4–18.1)	11.50 (8.5–14.5)	19.10 (15.0–23.2) [‡]
<i>Senegal</i>						
<i>Urban</i>						
	2003	37.1	35.2	21.80 (17.7–26.0)	18.10 (12.8–23.4)	25.30 (19.5–31.1) [‡]
<i>Rural</i>						
	2003	37.6	38.0	16.70 (13.4–20.1)	12.80 (8.9–16.8)	20.50 (15.3–25.6) [‡]
<i>Urban/rural</i>						
	2003	37.3	36.5	19.00 (16.4–21.6)	15.20 (11.9–18.5)	22.60 (18.8–26.5) [‡]
<i>West Africa</i>						
<i>Urban</i>						
	2003	35.8	35.0	18.00 (14.0–22.0)	14.00 (11.0–17.0)	21.00 (16.0–29.0)
<i>Rural</i>						
	2003	37.6	36.7	9.00 (6.0–13.0)	7.00 (5.0–10.0)	10.00 (7.0–16.0)
<i>Urban/rural</i>						
	2003	36.9	36.0	13.00 (9.0–18.0)	10.00 (7.0–13.0)	16.00 (11.0–22.0)

NEC, National Expert Committee; HoPiT, Health of Population in Transition Research Group–Cameroon; NR, not reported; NA, not applicable.

* $P < 0.05$, [†] $P < 0.001$, [‡] $P < 0.0001$ indicate P -value for statistical difference between men and women.^a Additional data from WHO InfoBase.

population.^{21,24,26,30,33,35} The increased prevalence of diabetes, particularly in urban West Africa, is also in line with the rise in obesity, so far mainly in women, observed in the past 15 years.¹⁶

The 4% prevalence of diabetes in urban West Africa is similar to rates in many European and other wealthy industrialized countries. It is, however, below the 9–14% prevalence reported for African-origin people in the Caribbean⁴⁹ and the UK,^{13,49} and the 13.2% prevalence in African Americans.¹⁴ It is also below the 16.3% prevalence reported among recent Ghanaian immigrants in Sydney.¹⁵ These figures from overseas signal what will be experienced in West Africa in the near future; indeed, over 9% of people in some urban centres already have diabetes.³¹

The finding of higher inactivity among women echoes observations elsewhere.^{13,50} Inactivity in urban West Africa seems to be higher than that reported (in World Health Surveys) from middle-income countries such as India in the same year. However, the rate of inactivity in West Africa does seem to be similar to rates from other African countries, including Mauritius, Zambia and Zimbabwe.⁴⁵

Given the high levels of inactivity and obesity¹⁶ among women, it was anticipated that the prevalence of diabetes would be higher among females. However, these data were from cross-sectional studies; a limited design not suitable for drawing causal inferences. Nonetheless, men seem to be more susceptible to the health

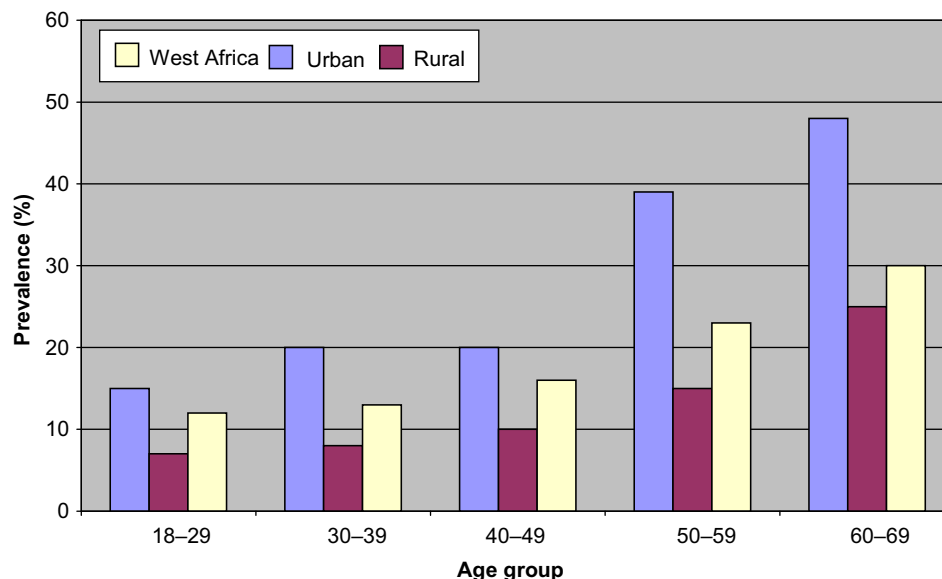


Figure 4. Age-specific inactivity in West Africa by residence (data from World Health Surveys).

consequences of obesity and inactivity.⁵¹ Moreover, although women may have a higher percentage of fat, their insulin sensitivity may be similar to their leaner male counterparts. Furthermore, obesity with a predominance of upper body fat (also called male-type obesity, abdominal obesity or truncal obesity) is more predictive of diabetes and other obesity-related morbidity than lower body fat obesity (i.e. peripheral obesity). In fact, the latter, which is more common in women, is even protective against metabolic disorders including type 2 diabetes.⁵²

Urban residents in West Africa have higher rates of obesity,¹⁶ physical inactivity and diabetes than rural residents. Remarkably, one study in this review even found a 0% prevalence of diabetes in some rural communities.³⁸ Meanwhile, at an annual growth rate of 4%, West Africa is experiencing the highest rate of urbanization in recent times. Twenty years ago, only one-quarter of West Africans lived in urban areas, but recent estimates indicate that 60% of the population will live in cities in the next 20 years. In fact, the greatest proportionate increase in urban population will be in West Africa.⁵³ Obesity, physical inactivity and diabetes will continue to increase unless policy makers, politicians, health promotion experts, researchers and all stakeholders urgently acknowledge and put in place workable policies (both research and healthcare programmes) to forestall further increases in these conditions.

Potential limitations of this review

This review sought to include data from all West African countries; however, most of the published studies came from urban Nigeria and urban Cameroon. This observation limits interpretation of the findings and underscores the need for further and more representative surveys in other parts of West Africa, including rural communities. The pooled estimates of this review were derived from a combination of studies conducted with different aims and among different population groups in West Africa; therefore, a considerable level of heterogeneity exists between the studies. However, to reduce the effect of heterogeneity, the authors meticulously combined studies that were deemed to be methodologically similar. In addition, the DerSimonian-Laird random effect model was used. This model explicitly accounts for variations within individual studies.^{54,55} Also, tests of heterogeneity in most

of the combined estimates reported in this review were not significant ($P > 0.05$). Other potential limitations relating to this review have been reported in the obesity data published elsewhere.¹⁶

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Competing interests

None declared.

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Appendix 1. List of excluded studies and reasons for exclusion

Number	Study reference	Reason(s) for exclusion
1	Abidoye et al., 2002	Sampled bank workers who are unrepresentative of the population. Also did not give explicit diagnostic criteria for diabetes
2	Rolfe et al., 1992	Used hospital-recorded cases
3	Ezenwaka et al., 1993	Sampled participants with family history of diabetes. Also did not report prevalence data
4	Osuntokun et al., 1971	Detailed descriptive study of already diagnosed diabetes
5	Ozumba et al., 2004	Retrospective case record study of diabetes in pregnancy
6	Amoah et al., 2002	Reported undiagnosed cases from already included study (see Reference 33)
7	Cooper et al., 1997	Reported data already included (see Reference 22)
8	Dodu, 1958	Reported incidence of diabetes in hospital-based patients
9	Mbanya et al., 1999	Data already reported in another study (see Reference 29)
10	Adedoyin et al., 2006	No prevalence data for any of the outcomes reported

Appendix 2. Full references for excluded studies

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